

Habiba Ehab

Operations Research, Data Analytics & Software Development Graduate
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PROFILE

Final year Decision Support Systems and Operations Research student at Cairo University, ranked among the top 5 in the department, graduating 2026, with a strong foundation in mathematical optimization and mixed integer linear programming, and hands-on experience across data analytics and business intelligence (Python, Power BI, Tableau) and web development (Angular, TypeScript, REST APIs). Led a top 5 graduation project from model design through delivery, and has applied structured problem solving and project management training through McKinsey Forward and a Microsoft backed program.

EDUCATION

B.Sc. Decision Support Systems & Operations Research

2022 - 2026

- Ranked among the top 5 students in the department | Graduation project selected as one of the top 5 projects in the department
- Relevant coursework: Linear, Nonlinear & Multi Objective Programming, Decision Theory, Systems Design, Data Structures, Database Management, Statistical Modeling, Machine Learning, Project Management

Orouba Language School (IGCSE & A-Level)

2022

- Graduated with high academic performance

TECHNICAL SKILLS

- Operations Research & Optimization: Linear, Nonlinear & Multi Objective Programming, Mixed Integer Linear Programming, Genetic Algorithms, Simulated Annealing, Queuing Theory, Decision Theory, Simulation
- Data Analytics & BI: Python (pandas, NumPy, Scikit-learn, XGBoost), Power BI, Tableau, SQL, Excel (Advanced)
- Web Development: JavaScript, TypeScript, Angular 19/20, NGRX, RxJS, PrimeNG, REST API Integration, HTML5, CSS3, SCSS, MySQL
- Languages & Tools: Python, C++ (university coursework), MATLAB, Flask, Git, GitHub, Visual Studio Code, Jira, Figma, CBC Solver, GAMS (coursework level), Microsoft Access (basic proficiency)
- Concepts: Agile/Scrum, OOP, Clean Code, State Management, Asynchronous Programming, CPM/PERT, Earned Value Management, Project Crashing

KEY PROJECTS

Operations Research & Decision Support

Schedify: Hybrid Work Scheduling Optimizer

Graduation Project, 2025 - 2026

- Led a team of 5 through model design, algorithm development, and delivery of a Mixed Integer Linear Programming based workforce scheduling system, with Genetic Algorithm and Simulated Annealing solvers benchmarked against an exact CBC solver for up to 1,470 employees
- Personally built the Angular frontend and managed integration and database schemas for a full stack web application (Flask backend) allowing non technical users to run the optimizer and export schedules

Weekly Meal Planning Optimizer & Two-Location Hospital Inventory Simulation

2024 - 2025

- Genetic Algorithm with fuzzy logic generating 7 day, 21 meal nutrition plans under nutritional and allergen constraints; and a Monte Carlo simulation modeling hospital inventory under stochastic demand and lead times

Additional Coursework Projects

2025 - 2026

- Multi Objective Site Selection (Eugene Public Library Case), Queuing Theory Simulation Toolkit (5 models), and an Ant Colony System metaheuristic for the Traveling Salesman Problem

Data Analytics & Business Intelligence

Hotel Booking Demand & Cancellation Modeling

2025

- Analyzed 50,000+ booking records; built and compared XGBoost (83%), Logistic Regression (81%), and Decision Trees (79%), then designed Power BI dashboards showing financial impact for pricing and inventory decisions

Student Performance Analysis & Ames Housing Price Prediction

2025

- Trained XGBoost and regression models on 60,000+ student records (87% accuracy) with a Power BI dashboard for non technical stakeholders; built an end to end regression pipeline on 2,900+ housing records (89% accuracy) with Tableau visualization

Road Accident Analytics Dashboard

2025

- Analyzed accident records across time, weather, and location to build a Tableau dashboard identifying risk trends for stakeholders

Apple Global Sales Dashboard

2025

- Cleaned and modeled global sales data with Power Query and Pivot Tables, then built a Power BI dashboard analyzing performance trends across 19 countries

Town-Team Sales Analysis

2025

- Built a native Excel sales dashboard tracking performance by item and category to support inventory and merchandising decisions

Web & Software Development

POS & CRM System (In Progress)

2026

- Implementing product search, filtering, and multi criteria sorting; integrating REST APIs for live data display and updates

AI Fitness App

2025

- Built frontend features in an Agile, Jira tracked sprint environment; implemented NGRX state management, token based authentication, and reactive form validation

Rose E-Commerce Platform

2024 - 2025

- Contributed to a multilingual e-commerce platform across Agile/Scrum sprints; built and maintained CRUD functionality with form validation and async error handling

TRAINING & PROFESSIONAL DEVELOPMENT

NTI Advanced Data Analytics

July - August 2025

- Intensive full time training (6 days a week, 6 hours a day, over 1 month) in Python, Power BI, and Tableau for data analysis and visualization

Elevate Tech Bootcamp: Frontend Advanced Track

2025

- Completed training in Angular development, clean architecture, micro frontends, advanced state management, and unit testing

McKinsey Forward Program

Completed June 2026

- 10 week global professional development program that covered: structured problem solving, communication frameworks, and adaptability skill building

Sprints x Microsoft Summer Camp: Project & Product Management Track

August 2025

- Coursework across the project lifecycle (initiation, planning, execution, monitoring and control, closure) and Agile & Scrum frameworks

Forage Virtual Job Simulations & L'Oreal Brandstorm 2026

2025 - 2026

- Deloitte (Data Analytics), Electronic Arts (Product Management), BCG (Introduction to Data for Decision Makers, completed; Introduction to Strategy Consulting, in progress); L'Oreal Brandstorm 2026 participant, pitched an original product concept

CERTIFICATIONS

- MathWorks MATLAB Optimization Onramp (May 2025) and MATLAB Onramp (October 2024)

LEADERSHIP & ACTIVITIES

- HR Committee Member, Cybersecurity Committee, Google Developer Student Clubs (GDSC), Cairo University (Sep 2023 - Aug 2024): Coordinated communication, attendance tracking, and task follow up. Recognized with the Best Member Award for reliability and consistent contribution
- Team Lead, Graduation Project (Schedify), Cairo University: Led a team of 5 through model design, algorithm development, and application delivery



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